Specific heat

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use mass, material and temperature change to calculate heating energy.

SUCCESS CRITERIA - I can...

1. Identify ΔT rather than final temperature alone.
2. Use Q=mcΔT with matching units.
3. Compare materials through specific heat.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “specific heat” and “heat” to explain your answer. Sentence starters are optional.

specific heat: energy per mass needed for a one-degree temperature change

heat: energy transferred because of a temperature difference

temperature: a measure of thermal state; it is not an energy amount

01 Warm-up: bring back an earlier idea

A device receives **50 J** and delivers **35 J** useful mechanical energy. **Find other energy** and the useful fraction.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $Q = mc\,\Delta T$; $\Delta T = T_f - T_i$.
No phase change. A temperature difference of **1°C** equals **1 K**.

Worked reasoning: A **2 kg** sample with c=**100 J**/(kg·°C) warms by **3°C**. Heat absorbed is Q=mc ΔT=**600 J**, with no phase change or work transfer in this calculation.

Your turn: Draw an energy arrow into the sample and **show which three factors produce 600 J**.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .

Initial T / phase: _____ Final T / phase: _____

Sample

Add an energy arrow; label mass, c or L as needed.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Two equal masses receive the same energy. Material A has a larger specific heat than B. Which has the smaller temperature rise?

Sentence starter: Both ... . They differ in ... because ... .



A: larger c



B: smaller c

Each outline represents the labeled sample or container.

Marzano check - circle: ① ② ③

04 Apply the physics

In a heating simulation, a **2 kg** sample has specific heat c = **100 J**/(kg·°C). Its temperature rises from **20°C** to **25°C** without a phase change.
1. **Label the temperatures** and energy arrow.
2. Find ΔT, then **the energy** absorbed.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Initial T / phase: _____ Final T / phase: _____

Sample

Add an energy arrow; label mass, c or L as needed.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

Why is multiplying by the final temperature instead of ΔT a mistake?

Sentence starter: The step I would change is … . I would change it because I found … .

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A pendulum loses **24 J** gravitational energy while descending. Neglect losses. **Find its kinetic-energy gain**.
Show the energy changes and transfers; **label units**.

Word help: pendulum = an object that swings about a fixed support

Sentence starter: I found … [unit]. This value means … in this situation.

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

07 Exit check

A **1 kg** sample with c=**200 J**/(kg·°C) warms by **3°C**.
1. **Find Q**.
2. **Label the energy arrow** and **identify what changes** in the sample.

Sentence starter: I found … [unit]. This value means … in this situation.

Initial T / phase: _____ Final T / phase: _____

Sample

Add an energy arrow; label mass, c or L as needed.

Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

Return to the worked example in problem **2**. The same **600 J** heats **1 kg** and **2 kg** samples of this material.
1. **Compare temperature rises**.
2. **Label the energy arrow** and **identify what changes** in the sample.

Sentence starter: Both … . They differ in … because … .

Initial T / phase: _____ Final T / phase: _____

Sample

Add an energy arrow; label mass, c or L as needed.

Marzano check - circle: ① ② ③